

Modeling the Economic Effects of the Section 232 Tariffs on Most Affected Industries

232 Model Release - 2019

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This model accompanies the USITC report, *Economic Impact of Section 232 and 301 Tariffs on U.S. Industries*, Inv. 332-591. The report includes a quantitative analysis of the economic impact of section 232 tariffs on U.S. prices, production and imports. The model inputs (ORANGE-shaded cells) include elasticities that describe responsiveness to price changes, and domestic production and imports data. Data sources are listed in the cell above the data input. The model inputs include the effects of the 232 tariffs. Then, the model simulation estimates what prices and quantities would have looked like if section 232 tariffs were not in place. Outputs are in the GREEN-shaded cells. The white cells are intermediate calculations.

Note: NAICS 336300 is an aggregate of NAICS 336330 and NAICS 336340.

```
In[ ]:= ClearAll[f];  
Clear["Global`*"];
```

1. Policy Changes

Existing Section 232 tariff on aluminum imports

```
In[ ]:= t232a0 = 0.10;
```

Counterfactual Section 232 tariff on aluminum imports

```
In[ ]:= t232a1 = 0.00;
```

Existing Section 232 tariff on steel imports

```
In[ ]:= t232s0 = 0.25;
```

Counterfactual Section 232 tariff on steel imports

```
In[ ]:= t232s1 = 0.00;
```

2. Downstream Industry Definitions

Number of downstream industries

```
In[ ]:= i = 33;
```

Downstream industry NAICS code

```
Inames = {2110, 312110, 3149, 3322, 3323, 3324, 3325, 3326, 3327, 3328,
          3329, 3331, 3332, 3334, 3335, 3339, 3351, 3352, 3353, 3359, 336212, 336214,
          336350, 336370, 336390, 336300, 336411, 3365, 3369, 3372, 3399, 3336, 3366};

In[ ]:= Table[NAICSCodes[t] = Inames[[t]], {t, 1, i}];
```

3. Parameter Inputs

3.1 Elasticity of Substitution

Source: USITC's econometric estimation using the trade cost method in Riker (2020).

Riker, David. "A Trade Cost Approach to Estimating the Elasticity of Substitution." *Economics Working Paper Series 2020-07-D*, U.S. International Trade Commission, July 2020. https://www.usitc.gov/publications/332/working_papers/ecwp_2017-07-d.pdf.

Steel

```
In[ ]:= sigmas = 1.493;
```

Aluminum

```
In[ ]:= sigmaa = 6.046;
```

Downstream industries

```
In[ ]:= sigmad = {5.42, 1.67, 6.33, 9.54, 4.54, 3.84, 5.24, 5.51, 4.22,
                  5, 5.74, 6.95, 7.80, 9.27, 8.90, 7.02, 5.34, 6.67, 9.68, 7.82, 3.06,
                  4.83, 6.48, 4.02, 4.28, 6.44, 3, 7.12, 7.16, 4.27, 6.78, 8.30, 7.51};
```

3.2 Domestic Supply Elasticities

Source: USITC staff estimate based on recent AD/CVD staff reports and data on domestic capacity utilization.

USITC, *Cut-to-Length Carbon-Quality Steel Plate from India, Indonesia, Italy, Japan, and Korea*, December 2011; USITC, *Hot-Rolled Flat-Rolled Carbon-Quality Steel Products from Brazil, Japan, and Russia*, June 2011; USITC, *Certain Hot-Rolled Steel Flat Products from Australia, Brazil, Japan, Korea, the Netherlands, Turkey, and the United Kingdom*, July 2016; USITC, *Cold-Rolled Steel Flat Products from China and Japan*, July 2016; Bown and Russ, "Biden and Europe Remove Trump's Steel and Aluminum Tariffs, but It's Not Free Trade," November 11, 2021.

Domestic steel

```
In[ ]:= eds = 2.5;
```

Domestic aluminum

```
In[ ]:= eda = 4;
```

3.3 Import Supply Elasticities

Source: USITC staff estimate using pass-through results from chapter 6 econometric analysis.

Steel subject imports

```
ln[ ]:= eis = 15;
```

Steel non-subject imports

```
ln[ ]:= ens = 15;
```

Aluminum subject imports

```
ln[ ]:= eia = 20;
```

Aluminum non-subject imports

```
ln[ ]:= ena = 20;
```

4. Data Inputs

4.1 Shipments of U.S. Domestic Steel to Each Downstream Industry

Source: USITC estimates. Total domestic production data were obtained from the U.S. Census Annual Survey of Manufactures. The use of steel by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values, see Appendix F of the report.

U.S. Census Bureau, "Annual Survey of Manufactures," accessed October 17, 2022. BEA, "Input-Output Accounts Data," accessed October 17, 2022.

```
ln[ ]:= vds0 = {6073176, 0, 596243, 614430, 9457785, 2931002, 266942, 2462163, 3922474,
1930097, 3352053, 4856617, 813004, 1052413, 1066724, 4388780, 494567,
757630, 990768, 1050089, 234145, 302122, 1051737, 4552422, 2315469, 971674,
1131216, 658859, 932512, 1098099, 903604, 755306, 266681}; (*in thousands*)
```

```
ln[ ]:= pds0 = 1;
```

```
ln[ ]:= vdstot0 = Total[vds0];
```

```
ln[ ]:= Table[qds0[t] = (vds0[[t]] / pds0), {t, 1, i}];
```

```
ln[ ]:= qdstot0 = vdstot0 / pds0;
```

4.2 Shipments of U.S. Domestic Aluminum to Each Downstream Industry

Source: USITC estimates. Total domestic production data were obtained from the U.S. Census Annual Survey of Manufactures. The use of aluminum by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values,

see Appendix F of the report.

U.S. Census Bureau, “Annual Survey of Manufactures,” accessed October 17, 2022. BEA, “Input-Output Accounts Data,” accessed October 17, 2022.

```
ln[ ]:= vda0 = {0, 6 699 546, 0, 297 283, 2 585 589, 5 940 037, 153 482, 44 495,
            1 359 083, 37 220, 1 701 368, 266 185, 304 032, 696 798, 493 395, 1 055 835,
            355 066, 179 485, 233 024, 704 557, 951 644, 163 164, 1 286 519, 2 345 187,
            1 192 817, 500 559, 57 371, 67 309, 557 868, 318 367, 318 023, 843 349, 634 523};
(*in thousands*)
```

```
ln[ ]:= pda0 = 1;
```

```
ln[ ]:= vdatot0 = Total[vda0];
```

```
ln[ ]:= Table[qda0[t] = vda0[[t]] / pda0, {t, 1, i}];
```

```
ln[ ]:= qdatot0 = vdatot0 / pda0;
```

4.3 Shipments of Steel Imports to Each Downstream Industry

Source: USITC estimates. Total import data were obtained from the USITC’s DataWeb. The use of steel by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values, see Appendix F of the report. USITC, “DataWeb/Census,” accessed September 28, 2022. BEA, “Input-Output Accounts Data,” accessed October 17, 2022.

Share of total steel imports that are subject to the Section 232 tariffs

```
ln[ ]:= shrs = 0.5094;
```

Subject imports

```
ln[ ]:= vis0 = shrs * {636 032, 0, 19 438, 224 949, 2 024 332, 586 587, 67 253, 262 695,
                    744 123, 714 819, 1 168 901, 1 649 314, 171 027, 210 347, 324 284, 1 319 352, 70 994,
                    164 583, 256 980, 213 529, 71 133, 89 265, 331 737, 936 515, 577 491, 241 855,
                    329 863, 164 583, 260 124, 245 479, 237 372, 248 966, 93 449}; (*in thousands*)
```

```
ln[ ]:= pis0 = 1;
```

```
ln[ ]:= vistot0 = Total[vis0];
```

```
ln[ ]:= Table[qis0[t] = vis0[[t]] / (pis0 (1 + t232s0)), {t, 1, i}];
```

```
ln[ ]:= qistot0 = vistot0 / (pis0 (1 + t232s0));
```

Non-subject imports

```
ln[ ]:= vns0 = (1 - shrs) * {636 032, 0, 19 438, 224 949, 2 024 332, 586 587, 67 253, 262 695,
                            744 123, 714 819, 1 168 901, 1 649 314, 171 027, 210 347, 324 284, 1 319 352, 70 994,
                            164 583, 256 980, 213 529, 71 133, 89 265, 331 737, 936 515, 577 491, 241 855,
                            329 863, 164 583, 260 124, 245 479, 237 372, 248 966, 93 449}; (*in thousands*)
```

```
ln[ ]:= pns0 = 1;
```

```
ln[ ]:= vnstot0 = Total[vns0];

ln[ ]:= Table[qns0[t] = vns0[[t]] / (pns0), {t, 1, i}];

ln[ ]:= qnstot0 = vnstot0 / (pns0);
```

4.4 Shipments of Aluminum Imports to Each Downstream Industry

Source: USITC estimates. Total import data were obtained from the USITC's DataWeb. The use of aluminum by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values, see Appendix F of the report. USITC, "DataWeb/Census," accessed September 28, 2022. BEA, "Input-Output Accounts Data," accessed October 17, 2022.

Share of total aluminum imports that are subject to the Section 232 tariffs

```
ln[ ]:= shra = 0.5728;
```

Subject imports

```
ln[ ]:= via0 = shra * {0, 1657509, 0, 105142, 637866, 2138373, 28100,
16213, 323303, 5418, 417261, 1356, 67477, 188096, 84216, 286448, 76046,
53049, 51305, 183770, 238278, 162766, 69366, 482447, 297495, 124592,
12674, 0, 241961, 130531, 89717, 29418, 145317}; (*in thousands*)
```

```
ln[ ]:= pia0 = 1;
```

```
ln[ ]:= viatot0 = Total[via0];
```

```
ln[ ]:= Table[qia0[t] = via0[[t]] / (pia0 (1 + t232a0)), {t, 1, i}];
```

```
ln[ ]:= qiatot0 = viatot0 / (pia0 (1 + t232a0));
```

Non-subject imports

```
ln[ ]:= vna0 = (1 - shra) * {0, 1657509, 0, 105142, 637866, 2138373, 28100,
16213, 323303, 5418, 417261, 1356, 67477, 188096, 84216, 286448, 76046,
53049, 51305, 183770, 238278, 162766, 69366, 482447, 297495, 124592,
12674, 0, 241961, 130531, 89717, 29418, 145317}; (*in thousands*)
```

```
ln[ ]:= pna0 = 1;
```

```
ln[ ]:= vnatot0 = Total[vna0];
```

```
ln[ ]:= Table[qna0[t] = vna0[[t]] / (pna0), {t, 1, i}];
```

```
ln[ ]:= qnatot0 = vnatot0 / (pna0);
```

4.5 Downstream data

Source: USITC estimates. Total domestic production data were obtained from the U.S. Census Annual Survey of Manufactures or estimated by the Commission's industry analysts using available production data from industry sources, if not available from the ASM. U.S. imports and exports data were obtained from USITC's DataWeb.

U.S. Census Bureau, "Annual Survey of Manufactures," accessed October 17, 2022. USITC, "DataWeb/Census," accessed September

5. Parameter Calibration

5.1 Supply parameters

$$\ln[\bullet] := \text{ads} = \text{qdstot0 pds0}^{-\text{eds}};$$

$$\ln[\bullet] := \text{ais} = \text{qistot0 pis0}^{-\text{eis}};$$

$$\ln[\bullet] := \text{ans} = \text{qnstot0 pns0}^{-\text{ens}};$$

$$\ln[\bullet] := \text{ada} = \text{qdatot0 pda0}^{-\text{eda}};$$

$$\ln[\bullet] := \text{aia} = \text{qiatot0 pia0}^{-\text{eia}};$$

$$\ln[\bullet] := \text{ana} = \text{qnatot0 pna0}^{-\text{ena}};$$

5.1 Demand parameters

$$\ln[\bullet] := \text{Table}[\text{bia}[t] = \text{If}[\text{vda0}[[t]] > 0, \left(\frac{\text{pda0}}{\text{pia0} (1 + t232a0)} \right)^{1-\text{sigmaa}} \left(\frac{\text{via0}[[t]]}{\text{vda0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\bullet] := \text{Table}[\text{bna}[t] = \text{If}[\text{vda0}[[t]] > 0, \left(\frac{\text{pda0}}{\text{pna0}} \right)^{1-\text{sigmaa}} \left(\frac{\text{vna0}[[t]]}{\text{vda0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\bullet] := \text{Table}[\text{za0}[t] = (\text{pda0}^{1-\text{sigmaa}} + \text{bia}[t] (\text{pia0} (1 + t232a0))^{1-\text{sigmaa}} + \text{bna}[t] \text{pna0}^{1-\text{sigmaa}})^{\frac{1}{1-\text{sigmaa}}}, \{t, 1, i\}];$$

$$\ln[\bullet] := \text{Table}[\text{bis}[t] = \text{If}[\text{vds0}[[t]] > 0, \left(\frac{\text{pds0}}{\text{pis0} (1 + t232s0)} \right)^{1-\text{sigmas}} \left(\frac{\text{vis0}[[t]]}{\text{vds0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\bullet] := \text{Table}[\text{bns}[t] = \text{If}[\text{vds0}[[t]] > 0, \left(\frac{\text{pds0}}{\text{pns0}} \right)^{1-\text{sigmas}} \left(\frac{\text{vns0}[[t]]}{\text{vds0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\bullet] := \text{Table}[\text{zs0}[t] = (\text{pds0}^{1-\text{sigmas}} + \text{bis}[t] (\text{pis0} (1 + t232s0))^{1-\text{sigmas}} + \text{bns}[t] \text{pns0}^{1-\text{sigmas}})^{\frac{1}{1-\text{sigmas}}}, \{t, 1, i\}];$$

Calibration of cost parameters

$$\ln[\bullet] := \text{Table}[\text{pd0}[t] = \text{w0}[[t]] + \text{bs0}[t] \times \text{zs0}[t] + \text{ba0}[t] \times \text{za0}[t], \{t, 1, i\}];$$

$$\ln[\bullet] := \text{Table}[\text{qd0}[t] = \text{vdd0}[t] / \text{pd0}[t], \{t, 1, i\}];$$

$$\ln[\bullet] := \text{Table}[\text{bi}[t] = \left(\frac{\text{pd0}[t]}{\text{pi0}[[t]]} \right)^{1-\text{sigmad}[[t]]} \left(\frac{\text{vi0}[[t]]}{\text{vdd0}[t]} \right), \{t, 1, i\}];$$

$$\ln[\bullet] := \text{Table}[\text{P0}[t] = (\text{pd0}[t]^{1-\text{sigmad}[[t]]} + \text{bi}[t] \text{pi0}[[t]]^{1-\text{sigmad}[[t]]})^{\frac{1}{1-\text{sigmad}[[t]]}}, \{t, 1, i\}];$$

$ln[] := \text{Table}[Cs[t] = css[t] ==$

$$\left(\frac{1}{vd0[t]} \right) \left(\left(pds0 \left(\frac{k0[t] \times bs0[t]}{pds0} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{zs0[t]}{pd0[t]} \right) \left(\frac{pds0}{zs0[t]} \right)^{1-sigmas} \right) + \right. \\ \left. \left(pis0 (1 + t232s0) \left(\frac{k0[t] \times bs0[t] \times bis[t]}{pis0 (1 + t232s0)} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \right. \right. \\ \left. \left(\frac{zs0[t]}{pd0[t]} \right) \left(\frac{pis0 (1 + t232s0)}{zs0[t]} \right)^{1-sigmas} \right) + \left(pns0 \left(\frac{k0[t] \times bs0[t] \times bns[t]}{pns0} \right) \right. \\ \left. \left. \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{zs0[t]}{pd0[t]} \right) \left(\frac{pns0}{zs0[t]} \right)^{1-sigmas} \right) \right), \{t, 1, i\}];$$

$ln[] := \text{Table}[Ca[t] =$

$$csa[t] == \left(\frac{1}{vd0[t]} \right) \left(\left(pda0 \left(\frac{k0[t] \times ba0[t]}{pda0} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{za0[t]}{pd0[t]} \right) \left(\frac{pda0}{za0[t]} \right)^{1-sigmaa} \right) + \right. \\ \left(pia0 (1 + t232a0) \left(\frac{k0[t] \times ba0[t] \times bia[t]}{pia0 (1 + t232a0)} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \right. \\ \left. \left(\frac{za0[t]}{pd0[t]} \right) \left(\frac{pia0 (1 + t232a0)}{za0[t]} \right)^{1-sigmaa} \right) + \left(pna0 \left(\frac{k0[t] \times ba0[t] \times bna[t]}{pna0} \right) \right. \\ \left. \left. \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{za0[t]}{pd0[t]} \right) \left(\frac{pna0}{za0[t]} \right)^{1-sigmaa} \right) \right), \{t, 1, i\}];$$

$ln[] := \text{Table}[Ck[t] = k0[t] == p0[t]^{1-sigmad[t]} pd0[t]^{sigmad[t]} qd0[t], \{t, 1, i\}];$

```

In[ ]:= sol = FindRoot[{Cs[1], Ca[1], Cs[2], Ca[2], Cs[3], Ca[3], Cs[4], Ca[4], Cs[5], Ca[5],
Cs[6], Ca[6], Cs[7], Ca[7], Cs[8], Ca[8], Cs[9], Ca[9], Cs[10], Ca[10], Cs[11],
Ca[11], Cs[12], Ca[12], Cs[13], Ca[13], Cs[14], Ca[14], Cs[15], Ca[15], Cs[16],
Ca[16], Cs[17], Ca[17], Cs[18], Ca[18], Cs[19], Ca[19], Cs[20], Ca[20], Cs[21],
Ca[21], Cs[22], Ca[22], Cs[23], Ca[23], Cs[24], Ca[24], Cs[25], Ca[25], Cs[26],
Ca[26], Cs[27], Ca[27], Cs[28], Ca[28], Cs[29], Ca[29], Cs[30], Ca[30], Cs[31],
Ca[31], Cs[32], Ca[32], Cs[33], Ca[33], Ck[1], Ck[2], Ck[3], Ck[4], Ck[5],
Ck[6], Ck[7], Ck[8], Ck[9], Ck[10], Ck[11], Ck[12], Ck[13], Ck[14], Ck[15],
Ck[16], Ck[17], Ck[18], Ck[19], Ck[20], Ck[21], Ck[22], Ck[23], Ck[24],
Ck[25], Ck[26], Ck[27], Ck[28], Ck[29], Ck[30], Ck[31], Ck[32], Ck[33]},
{bs0[1], css[1]}, {ba0[1], csa[1]}, {k0[1], vdd0[1] + vi0[1]}, {bs0[2], css[2]},
{ba0[2], csa[2]}, {k0[2], vdd0[2] + vi0[2]}, {bs0[3], css[3]},
{ba0[3], csa[3]}, {k0[3], vdd0[3] + vi0[3]}, {bs0[4], css[4]},
{ba0[4], csa[4]}, {k0[4], vdd0[4] + vi0[4]}, {bs0[5], css[5]},
{ba0[5], csa[5]}, {k0[5], vdd0[5] + vi0[5]}, {bs0[6], css[6]},
{ba0[6], csa[6]}, {k0[6], vdd0[6] + vi0[6]}, {bs0[7], css[7]},
{ba0[7], csa[7]}, {k0[7], vdd0[7] + vi0[7]}, {bs0[8], css[8]}, {ba0[8], csa[8]},
{k0[8], vdd0[8] + vi0[8]}, {bs0[9], css[9]}, {ba0[9], csa[9]}, {k0[9], vdd0[9] + vi0[9]},
{bs0[10], css[10]}, {ba0[10], csa[10]}, {k0[10], vdd0[10] + vi0[10]},
{bs0[11], css[11]}, {ba0[11], csa[11]}, {k0[11], vdd0[11] + vi0[11]},
{bs0[12], css[12]}, {ba0[12], csa[12]}, {k0[12], vdd0[12] + vi0[12]},
{bs0[13], css[13]}, {ba0[13], csa[13]}, {k0[13], vdd0[13] + vi0[13]},
{bs0[14], css[14]}, {ba0[14], csa[14]}, {k0[14], vdd0[14] + vi0[14]},
{bs0[15], css[15]}, {ba0[15], csa[15]}, {k0[15], vdd0[15] + vi0[15]},
{bs0[16], css[16]}, {ba0[16], csa[16]}, {k0[16], vdd0[16] + vi0[16]},
{bs0[17], css[17]}, {ba0[17], csa[17]}, {k0[17], vdd0[17] + vi0[17]},
{bs0[18], css[18]}, {ba0[18], csa[18]}, {k0[18], vdd0[18] + vi0[18]},
{bs0[19], css[19]}, {ba0[19], csa[19]}, {k0[19], vdd0[19] + vi0[19]},
{bs0[20], css[20]}, {ba0[20], csa[20]}, {k0[20], vdd0[20] + vi0[20]},
{bs0[21], css[21]}, {ba0[21], csa[21]}, {k0[21], vdd0[21] + vi0[21]},
{bs0[22], css[22]}, {ba0[22], csa[22]}, {k0[22], vdd0[22] + vi0[22]},
{bs0[23], css[23]}, {ba0[23], csa[23]}, {k0[23], vdd0[23] + vi0[23]},
{bs0[24], css[24]}, {ba0[24], csa[24]}, {k0[24], vdd0[24] + vi0[24]},
{bs0[25], css[25]}, {ba0[25], csa[25]}, {k0[25], vdd0[25] + vi0[25]},
{bs0[26], css[26]}, {ba0[26], csa[26]}, {k0[26], vdd0[26] + vi0[26]},
{bs0[27], css[27]}, {ba0[27], csa[27]}, {k0[27], vdd0[27] + vi0[27]},
{bs0[28], css[28]}, {ba0[28], csa[28]}, {k0[28], vdd0[28] + vi0[28]},
{bs0[29], css[29]}, {ba0[29], csa[29]}, {k0[29], vdd0[29] + vi0[29]},
{bs0[30], css[30]}, {ba0[30], csa[30]}, {k0[30], vdd0[30] + vi0[30]},
{bs0[31], css[31]}, {ba0[31], csa[31]}, {k0[31], vdd0[31] + vi0[31]},
{bs0[32], css[32]}, {ba0[32], csa[32]}, {k0[32], vdd0[32] + vi0[32]},
{bs0[33], css[33]}, {ba0[33], csa[33]}, {k0[33], vdd0[33] + vi0[33]}}];

```

$\text{In}[*]:= \text{Table}[\text{bs}[t] = \text{bs0}[t] /. \text{sol}, \{t, 1, i\}]$

$\text{Out}[*]= \{0.0327275, 0., 0.0773027, 0.256481, 0.202118, 0.231798, 0.0928061, 0.634178, 0.108873, 0.194404, 0.184677, 0.182911, 0.105137, 0.0493311, 0.0989821, 0.129855, 0.0667974, 0.0879506, 0.0764952, 0.0663074, 0.0565398, 0.04054, 0.0637517, 0.293287, 0.0989565, 0.180634, 0.0252285, 0.15063, 0.125599, 0.0829241, 0.0433402, 0.0846409, 0.0201617\}$

$\text{In}[*]:= \text{Table}[\text{ba}[t] = \text{ba0}[t] /. \text{sol}, \{t, 1, i\}]$

$\text{Out}[*]= \{0., 0.243132, 0., 0.0693468, 0.0399967, 0.390792, 0.0330508, 0.0122335, 0.0287866, 0.00169916, 0.0492705, 0.00416103, 0.0280402, 0.0250463, 0.0247513, 0.01879, 0.0403097, 0.0156678, 0.011358, 0.0335277, 0.134498, 0.0229041, 0.0362039, 0.107283, 0.0339149, 0.0619531, 0.000748794, 0.00783285, 0.0549213, 0.0196973, 0.0101371, 0.0415535, 0.0247276\}$

$\text{In}[*]:= \text{Table}[\text{k}[t] = \text{k0}[t] /. \text{sol}, \{t, 1, i\}]$

$\text{Out}[*]= \{3.81325 \times 10^8, 4.80823 \times 10^7, 1.67664 \times 10^7, 1.66072 \times 10^7, 1.05547 \times 10^8, 3.84868 \times 10^7, 1.44482 \times 10^7, 1.14486 \times 10^7, 7.31371 \times 10^7, 2.84661 \times 10^7, 8.7797 \times 10^7, 1.08037 \times 10^8, 4.22632 \times 10^7, 5.56845 \times 10^7, 4.09073 \times 10^7, 1.37585 \times 10^8, 2.60906 \times 10^7, 4.77374 \times 10^7, 6.26561 \times 10^7, 7.17791 \times 10^7, 1.32054 \times 10^7, 1.77618 \times 10^7, 4.96712 \times 10^7, 3.63125 \times 10^7, 8.69366 \times 10^7, 3.32607 \times 10^7, 1.13305 \times 10^8, 1.09148 \times 10^7, 2.49233 \times 10^7, 3.42137 \times 10^7, 1.2918 \times 10^8, 4.82623 \times 10^7, 3.61837 \times 10^7\}$

$\text{In}[*]:= \text{Table}[\text{pd0}[t] = (\text{w0}[[t]] + \text{bs}[t] \times \text{zs0}[t] + \text{ba}[t] \times \text{za0}[t]), \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{qd0}[t] = \text{vdd0}[t] / \text{pd0}[t], \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{bi}[t] = \left(\frac{\text{pd0}[t]}{\text{pi0}[[t]]} \right)^{1-\text{sigmad}[[t]]} \left(\frac{\text{vi0}[[t]]}{\text{vdd0}[t]} \right), \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{Pindex0}[t] = (\text{pd0}[t]^{1-\text{sigmad}[[t]]} + \text{bi}[t] \text{pi0}[[t]]^{1-\text{sigmad}[[t]]})^{\frac{1}{1-\text{sigmad}[[t]]}}, \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{za0}[t] = \text{If}[\text{adinp}[t] + \text{aiinp}[t] + \text{aninp}[t] > 0, (\text{pda0}^{1-\text{sigmaa}} + \text{bia}[t] (\text{pia0} (1 + \text{t232a0}))^{1-\text{sigmaa}} + \text{bna}[t] \text{pna0}^{1-\text{sigmaa}})^{\frac{1}{1-\text{sigmaa}}}, 0], \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{zs0}[t] = \text{If}[\text{sdinp}[t] + \text{siinp}[t] + \text{sninp}[t] > 0, (\text{pds0}^{1-\text{sigmas}} + \text{bis}[t] (\text{pis0} (1 + \text{t232s0}))^{1-\text{sigmas}} + \text{bns}[t] \text{pns0}^{1-\text{sigmas}})^{\frac{1}{1-\text{sigmas}}}, 0], \{t, 1, i\}];$

6. New Equilibrium

$\text{In}[*]:= \mathbf{w} = \mathbf{w0};$

$\text{In}[*]:= \mathbf{pi} = \mathbf{pi0};$

$\text{In}[*]:= \text{Table}[\text{za}[t] = (\text{pda}^{1-\text{sigmaa}} + \text{bia}[t] (\text{pia} (1 + \text{t232a1}))^{1-\text{sigmaa}} + \text{bna}[t] \text{pna}^{1-\text{sigmaa}})^{\frac{1}{1-\text{sigmaa}}}, \{t, 1, i\}];$

```
ln[ ]:= Table[
  zs[t] = (pds1-sigmas + bis[t] (pis (1 + t232s1))1-sigmas + bns[t] pns1-sigmas) $\frac{1}{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Table[pd[t] = (w[[t]] + bs[t] × zs[t] + ba[t] × za[t]), {t, 1, i}];
```

```
ln[ ]:= Table[P[t] = (pd[t]1-sigmad[[t]] + bi[t] pi[[t]]1-sigmad[[t]]) $\frac{1}{1-sigmad[[t]]}$ , {t, 1, i}];
```

Equilibrium equations

```
ln[ ]:= Table[qda[t] =  $\left(\frac{k[t] \times ba[t]}{pda}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{za[t]}{pd[t]}\right) \left(\frac{pda}{za[t]}\right)^{1-sigmaa}$ , {t, 1, i}];
```

```
ln[ ]:= Eda = ada pdaeda == Sum[qda[t], {t, 1, i}];
```

```
ln[ ]:= Table[qia[t] =
   $\left(\frac{k[t] \times ba[t] \times bia[t]}{pia (1 + t232a1)}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{za[t]}{pd[t]}\right) \left(\frac{pia (1 + t232a1)}{za[t]}\right)^{1-sigmaa}$ , {t, 1, i}];
```

```
ln[ ]:= Eia = aia piaeia == Sum[qia[t], {t, 1, i}];
```

```
ln[ ]:= Table[qna[t] =  $\left(\frac{k[t] \times ba[t] \times bna[t]}{pna}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{za[t]}{pd[t]}\right) \left(\frac{pna}{za[t]}\right)^{1-sigmaa}$ , {t, 1, i}];
```

```
ln[ ]:= Ena = ana pnaena == Sum[qna[t], {t, 1, i}];
```

```
ln[ ]:= Table[qds[t] =  $\left(\frac{k[t] \times bs[t]}{pds}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs[t]}{pd[t]}\right) \left(\frac{pds}{zs[t]}\right)^{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Eds = ads pdseds == Sum[qds[t], {t, 1, i}];
```

```
ln[ ]:= Table[qis[t] =
   $\left(\frac{k[t] \times bs[t] \times bis[t]}{pis (1 + t232s1)}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs[t]}{pd[t]}\right) \left(\frac{pis (1 + t232s1)}{zs[t]}\right)^{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Eis = ais piseis == Sum[qis[t], {t, 1, i}];
```

```
ln[ ]:= Table[qns[t] =  $\left(\frac{k[t] \times bs[t] \times bns[t]}{pns}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs[t]}{pd[t]}\right) \left(\frac{pns}{zs[t]}\right)^{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Ens = ans pnsens == Sum[qns[t], {t, 1, i}];
```

```
ln[ ]:= eqm = FindRoot[{Eda, Eia, Ena, Eds, Eis, Ens}, {pda, pda0},
  {pia, pia0}, {pna, pna0}, {pds, pds0}, {pis, pis0}, {pns, pns0}]
```

```
Out[ ]:= {pda → 0.989146, pia → 1.01787, pna → 0.995601, pds → 0.991353, pis → 1.01813, pns → 0.997767}
```

```
ln[ ]:= pda1 = pda /. eqm;
```

```
ln[ ]:= pia1 = pia /. eqm;
```

```
ln[ ]:= pna1 = pna /. eqm;
```

```

ln[ ]:= pds1 = pds /. eqm;

ln[ ]:= pis1 = pis /. eqm;

ln[ ]:= pns1 = pns /. eqm;

ln[ ]:= Table[za1[t] = If[adinp[t] + aiinp[t] + aninp[t] > 0,
  (pda11-sigmaa + bia[t] (pia1 (1 + t232a1))1-sigmaa + bna[t] pna11-sigmaa) $\frac{1}{1-sigmaa}$ , 0], {t, 1, i}];

ln[ ]:= Table[zs1[t] = If[sdinp[t] + siinp[t] × sninp[t] > 0,
  (pds11-sigmas + bis[t] (pis1 (1 + t232s1))1-sigmas + bns[t] pns11-sigmas) $\frac{1}{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[pd1[t] = (w[[t]] + bs[t] × zs1[t] + ba[t] × za1[t]), {t, 1, i}];

ln[ ]:= Table[Pindex1[t] = (pd1[t]1-sigmad[[t]] + bi[t] pi[[t]]1-sigmad[[t]]) $\frac{1}{1-sigmad[[t]]}$ , {t, 1, i}];

ln[ ]:= Table[qd1[t] = k[t] Pindex1[t]sigmad[[t]]-1 pd1[t]-sigmad[[t]], {t, 1, i}];

ln[ ]:= qdatot1 = ada pda1eda;

ln[ ]:= qiatot1 = aia pia1eia;

ln[ ]:= qnatot1 = ana pna1ena;

ln[ ]:= qdstot1 = ads pds1eds;

ln[ ]:= qistot1 = ais pis1eis;

ln[ ]:= qnstot1 = ans pns1ens;

ln[ ]:= Table[qda1[t] = If[adinp[t] > 0,
  ( (  $\frac{k[t] \times ba[t]}{pda1}$  ) (  $\frac{pd1[t]}{Pindex1[t]}$  )1-sigmad[[t]] (  $\frac{za1[t]}{pd1[t]}$  ) (  $\frac{pda1}{za1[t]}$  )1-sigmaa ), 0], {t, 1, i}];

ln[ ]:= Table[qia1[t] = If[aiinp[t] > 0, (  $\frac{k[t] \times ba[t] \times bia[t]}{pia1 (1 + t232a1)}$  )
  (  $\frac{pd1[t]}{Pindex1[t]}$  )1-sigmad[[t]] (  $\frac{za1[t]}{pd1[t]}$  ) (  $\frac{pia1 (1 + t232a1)}{za1[t]}$  )1-sigmaa, 0], {t, 1, i}];

ln[ ]:= Table[qna1[t] = If[aninp[t] > 0,
  (  $\frac{k[t] \times ba[t] \times bna[t]}{pna1}$  ) (  $\frac{pd1[t]}{Pindex1[t]}$  )1-sigmad[[t]] (  $\frac{za1[t]}{pd1[t]}$  ) (  $\frac{pna1}{za1[t]}$  )1-sigmaa, 0], {t, 1, i}];

ln[ ]:= Table[vda1[t] = If[adinp[t] > 0,
  pda1 * (  $\frac{k[t] \times ba[t]}{pda1}$  ) (  $\frac{pd1[t]}{Pindex1[t]}$  )1-sigmad[[t]] (  $\frac{za1[t]}{pd1[t]}$  ) (  $\frac{pda1}{za1[t]}$  )1-sigmaa, 0], {t, 1, i}];

```

```

ln[ ]:= Table[via1[t] = If[aiinp[t] > 0, pia1 (1 + t232a1) *  $\left(\frac{k[t] \times ba[t] \times bia[t]}{pia1 (1 + t232a1)}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{za1[t]}{pd1[t]}\right) \left(\frac{pia1 (1 + t232a1)}{za1[t]}\right)^{1-sigmaa}$ , 0], {t, 1, i}];

ln[ ]:= Table[vna1[t] = If[aninp[t] > 0, pna1 *  $\left(\frac{k[t] \times ba[t] \times bna[t]}{pna1}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{za1[t]}{pd1[t]}\right) \left(\frac{pna1}{za1[t]}\right)^{1-sigmaa}$ , 0], {t, 1, i}];

ln[ ]:= Table[qds1[t] = If[sdinp[t] > 0,
 $\left(\left(\frac{k[t] \times bs[t]}{pds1}\right) \left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pds1}{zs1[t]}\right)^{1-sigma}$ ), 0], {t, 1, i}];

ln[ ]:= Table[qis1[t] = If[siinp[t] > 0,  $\left(\frac{k[t] \times bs[t] \times bis[t]}{pis1 (1 + t232s1)}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pis1 (1 + t232s1)}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[qns1[t] = If[sninp[t] > 0,
 $\left(\frac{k[t] \times bs[t] \times bns[t]}{pns1}\right) \left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pns1}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[vds1[t] = If[sdinp[t] > 0,
pds1 *  $\left(\left(\frac{k[t] \times bs[t]}{pds1}\right) \left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pds1}{zs1[t]}\right)^{1-sigma}$ ), 0], {t, 1, i}];

ln[ ]:= Table[vis1[t] = If[siinp[t] > 0, pis1 (1 + t232s1) *  $\left(\frac{k[t] \times bs[t] \times bis[t]}{pis1 (1 + t232s1)}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pis1 (1 + t232s1)}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[vns1[t] = If[sninp[t] > 0, pns1 *  $\left(\frac{k[t] \times bs[t] \times bns[t]}{pns1}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pns1}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

```

7. Results

Aluminum

$$\ln[\circ] := 100 \left(\frac{\text{pda0} - \text{pda1}}{\text{pda1}} \right)$$

$$\text{Out}[\circ] = 1.09734$$

$$\ln[\circ] := 100 \left(\frac{\text{pia0} - \text{pia1}}{\text{pia1}} \right)$$

$$\text{Out}[\circ] = -1.75595$$

$$\ln[\circ] := 100 \frac{\text{pia0} (1 + \text{t232a0}) - \text{pia1} (1 + \text{t232a1})}{\text{pia1} (1 + \text{t232a1})}$$

$$\text{Out}[\circ] = 8.06845$$

$$\ln[\circ] := 100 \left(\frac{\text{pna0} - \text{pna1}}{\text{pna1}} \right)$$

$$\text{Out}[\circ] = 0.441835$$

$$\ln[\circ] := \text{qdachange} = 100 \left(\frac{\text{qdatot0} - \text{qdatot1}}{\text{qdatot1}} \right)$$

$$\text{Out}[\circ] = 4.46212$$

$$\ln[\circ] := 100 \left(\frac{\text{qiatot0} - \text{qiatot1}}{\text{qiatot1}} \right)$$

$$\text{Out}[\circ] = -29.8343$$

$$\ln[\circ] := 100 \left(\frac{\text{qnatot0} - \text{qnatot1}}{\text{qnatot1}} \right)$$

$$\text{Out}[\circ] = 9.21764$$

Steel

$$\ln[\circ] := 100 \left(\frac{\text{pds0} - \text{pds1}}{\text{pds1}} \right)$$

$$\text{Out}[\circ] = 0.872268$$

$$\ln[\bullet] := 100 \left(\frac{\text{pis0} - \text{pis1}}{\text{pis1}} \right)$$

Out[•]= -1.78042

$$\ln[\bullet] := 100 \frac{\text{pis0} (1 + t232s0) - \text{pis1} (1 + t232s1)}{\text{pis1} (1 + t232s1)}$$

Out[•]= 22.7745

$$\ln[\bullet] := 100 \left(\frac{\text{pns0} - \text{pns1}}{\text{pns1}} \right)$$

Out[•]= 0.223756

$$\ln[\bullet] := \text{qdschange} = 100 \left(\frac{\text{qdstot0} - \text{qdstot1}}{\text{qdstot1}} \right)$$

Out[•]= 2.19496

$$\ln[\bullet] := 100 \left(\frac{\text{qistot0} - \text{qistot1}}{\text{qistot1}} \right)$$

Out[•]= -23.6215

$$\ln[\bullet] := 100 \left(\frac{\text{qnstot0} - \text{qnstot1}}{\text{qnstot1}} \right)$$

Out[•]= 3.40942

Downstream

$\ln[\bullet] := \text{Inames}(*\text{naics codes}*)$

Out[•]= {2110, 312110, 3149, 3322, 3323, 3324, 3325, 3326, 3327, 3328, 3329,
3331, 3332, 3334, 3335, 3339, 3351, 3352, 3353, 3359, 336212, 336214,
336350, 336370, 336390, 336300, 336411, 3365, 3369, 3372, 3399, 3336, 3366}

$$\ln[\bullet] := \text{Table} \left[\text{pddchangex} [t] = 100 \left(\frac{\text{pd0}[t] - \text{pd1}[t]}{\text{pd1}[t]} \right), \{t, 1, i\} \right]$$

Out[•]= {0.0473168, 0.360318, 0.0799661, 0.518613, 0.367789, 0.792549, 0.205682, 0.639522,
0.217702, 0.332516, 0.379936, 0.314107, 0.214328, 0.125809, 0.207712, 0.252843, 0.163884,
0.173649, 0.150964, 0.165635, 0.302737, 0.130748, 0.154277, 0.539822, 0.220974,
0.376772, 0.0472557, 0.256188, 0.311089, 0.176998, 0.0950399, 0.191856, 0.0789911}

```
In[ ]:= Table[100  $\left( \frac{\text{Pindex0}[t] - \text{Pindex1}[t]}{\text{Pindex1}[t]} \right)$ , {t, 1, i}]
```

```
Out[ ]:= {0.0319737, 0.331726, 0.0435112, 0.23371, 0.344723, 0.691869,  
0.0883721, 0.451894, 0.200416, 0.332516, 0.223731, 0.207365, 0.0770934,  
0.088666, 0.132593, 0.151018, 0.0761851, 0.0611913, 0.0666488,  
0.0691728, 0.246358, 0.125436, 0.126127, 0.530304, 0.128009, 0.141018,  
0.0412278, 0.222793, 0.220613, 0.135543, 0.0322742, 0.0918148, 0.0742179}
```

```
In[ ]:= Table[qddchangex[t] = 100  $\left( \frac{\text{qd0}[t] - \text{qd1}[t]}{\text{qd1}[t]} \right)$ , {t, 1, i}]
```

```
Out[ ]:= {-0.115029, -0.378045, -0.273744, -2.8984, -0.447475, -1.06751,  
-0.699677, -1.46821, -0.272637, -0.331414, -1.11118, -0.94261, -1.1394,  
-0.431642, -0.796735, -0.860554, -0.542432, -0.807114, -0.878035, -0.819221,  
-0.417229, -0.150868, -0.307726, -0.565357, -0.523748, -1.64163, -0.0592774,  
-0.458691, -0.862721, -0.311699, -0.456504, -0.916715, -0.109949}
```

```
In[ ]:= Table[vddchangex[t] = (pd0[t] × qd0[t] - pd1[t] × qd1[t]), {t, 1, i}]
```

```
Out[ ]:= {-174691., -8452.15, -17718.6, -181658., -80501.4, -95405.5, -30849.6, -68146.9,  
-37401.2, 0., -381940., -452046., -141754., -120442., -154760., -502972., -46128.6,  
-107197., -202310., -197070., -12449.9, -3462.57, -62562.7, -10199.5, -153369.,  
-159430., -11912.5, -19356.1, -98289.9, -35468.4, -159094., -168519., -10556.3}
```

```
In[ ]:= Table[vdachange[t] = If[ vda1[t] > 0, vda0[t] - vda1[t], 0], {t, 1, i}];
```

```
In[ ]:= Table[vdschange[t] = If[ vds1[t] > 0, vds0[t] - vds1[t], 0], {t, 1, i}];
```

```
In[ ]:= domesticchange = Prepend[  
Table[{Inames[t], qddchangex[t], pddchangex[t],  $\frac{100 (\text{pd0}[t] \times \text{qd0}[t] - \text{pd1}[t] \times \text{qd1}[t])}{\text{pd1}[t] \times \text{qd1}[t]}$ ,  
vddchangex[t], vdachange[t], vdschange[t]}, {t, 1, i}],  
{"NAICS", "Percent change domestic quantity", "Percent change domestic price",  
"Percent change in value", "Change in value",  
"Change in domestic aluminum", "Change in domestic steel"}];
```

In[]:= Grid[domesticchange, Frame → All]

Out[]:=

NAICS	Percent change domestic quantity	Percent change domestic price	Percent change in value	Change in value	Change in domestic aluminum	Change in domestic steel
2110	-0.115029	0.0473168	-0.0677669	-174 691.	0	131 725.
312 110	-0.378045	0.360318	-0.0190891	-8452.15	372 675.	0
3149	-0.273744	0.0799661	-0.193997	-17 718.6	0	6372.39
3322	-2.8984	0.518613	-2.39482	-181 658.	13 672.	12 039.7
3323	-0.447475	0.367789	-0.0813319	-80 501.4	141 851.	287 918.
3324	-1.06751	0.792549	-0.283421	-95 405.5	382 712.	67 295.2
3325	-0.699677	0.205682	-0.495434	-30 849.6	6484.38	8442.72
3326	-1.46821	0.639522	-0.838079	-68 146.9	2721.58	20 917.1
3327	-0.272637	0.217702	-0.055529	-37 401.2	74 998.9	116 413.
3328	-0.331414	0.332516	0.	0.	1457.17	87 183.6
3329	-1.11118	0.379936	-0.735464	-381 940.	82 173.9	120 497.
3331	-0.94261	0.314107	-0.631464	-452 046.	730.925	179 007.
3332	-1.1394	0.214328	-0.927511	-141 754.	13 500.6	18 933.4
3334	-0.431642	0.125809	-0.306376	-120 442.	40 716.6	30 701.9
3335	-0.796735	0.207712	-0.590678	-154 760.	19 323.6	37 679.8
3339	-0.860554	0.252843	-0.609886	-502 972.	57 602.5	151 029.
3351	-0.542432	0.163884	-0.379437	-46 128.6	17 364.4	10 902.6
3352	-0.807114	0.173649	-0.634866	-107 197.	10 497.7	20 641.3
3353	-0.878035	0.150964	-0.728396	-202 310.	10 869.1	30 291.9
3359	-0.819221	0.165635	-0.654943	-197 070.	37 641.3	27 021.7
336 212	-0.417229	0.302737	-0.115755	-12 449.9	53 008.1	9127.49
336 214	-0.150868	0.130748	-0.0203174	-3462.57	20 144.9	12 336.7
336 350	-0.307726	0.154277	-0.153924	-62 562.7	26 682.8	43 137.5
336 370	-0.565357	0.539822	-0.0285868	-10 199.5	110 969.	129 562.
336 390	-0.523748	0.220974	-0.303931	-153 369.	65 060.7	76 651.8
336 300	-1.64163	0.376772	-1.27105	-159 430.	21 885.7	21 442.7
336 411	-0.0592774	0.0472557	-0.0120497	-11 912.5	3130.8	46 809.6
3365	-0.458691	0.256188	-0.203679	-19 356.1	423.793	22 251.6
3369	-0.862721	0.311089	-0.554316	-98 289.9	41 788.1	30 303.2
3372	-0.311699	0.176998	-0.135253	-35 468.4	24 641.1	35 905.2
3399	-0.456504	0.0950399	-0.361898	-159 094.	19 059.7	31 610.6
3336	-0.916715	0.191856	-0.726618	-168 519.	8699.5	27 416.8
3366	-0.109949	0.0789911	-0.0310451	-10 556.3	35 126.3	12 199.2